

LINUX ESSENTIALS



```
$ whoami
```

```
linux_student
```

```
$ echo $LECTURE
```

```
"vi + Command Line"
```

```
$ _
```

The vi Editor & the Command Line

Core commands • 3 vi modes • 1 fifty-minute session

What we'll cover in 50 minutes

01 The vi Editor

`modes, movement, save & quit`

02 Navigation & Directories

`pwd · cd · mkdir · rmdir · ls`

03 Copying, Moving & Removing

`cp · mv · rm`

04 Viewing & Inspecting Files

`cat · more · wc · od`

05 Printing & Archiving

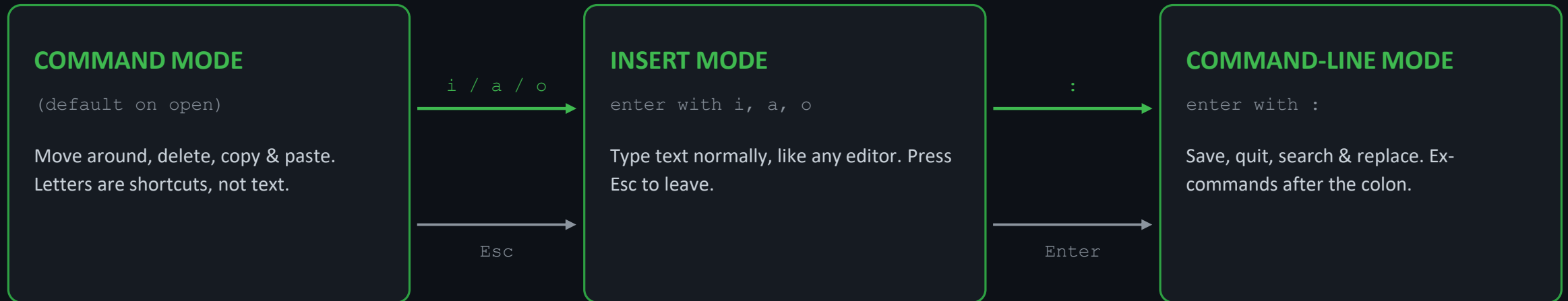
`lp · tar · gzip`

06 Live Practice

`run it together on Ubuntu`

One Editor, Three Modes

Everything in vi confuses beginners exactly once: it's modal. Know your mode, know your power.



Rule of thumb: press `Esc` if you're ever unsure which mode you're in.

Essential vi Commands

MOVEMENT (command mode)

h j k l left / down / up / right

0 / \$ start / end of line

gg / G top / bottom of file

w / b next / previous word

EDITING (command mode)

i / a insert before / after cursor

o open new line below

x delete character

dd / yy delete / copy (yank) line

p paste after cursor

u undo

SAVE & QUIT (command-line mode)

:w save (write)

:q quit

:wq or ZZ save and quit

:q! quit without saving

Navigation & Directories

Find out where you are, move around, and manage folders.



`pwd`

Prints the full path of your current (working) directory.



`cd`

Changes directory. `cd ..` goes up one level, `cd ~` goes home, `cd -` goes back.



`ls`

Lists directory contents. Try `ls -la` for hidden files and details.



`mkdir`

Creates a new directory.



`rmdir`

Removes an empty directory.

Copying, Moving & Removing

The three commands you'll use constantly to manage files.



cp

Copies files or directories. Use -r to copy a folder and everything inside it.



mv

Moves a file to a new location, or renames it if the destination is a new filename.



rm

Deletes files permanently. Use -r for folders, -f to force — there's no recycle bin!

Viewing & Inspecting Files

Look inside a file without opening a full editor.



cat

Prints an entire file to the screen. Also joins files: `cat a.txt b.txt > both.txt`



more

Shows a file one screen at a time — Space to page forward, q to quit. Good for long files.



wc

Counts lines, words, and characters. `wc -l file` counts just the lines.



od

Octal dump — shows a file's raw bytes. Useful for binary or non-printable content.

Printing & Archiving

Send output to a printer, and bundle up files to share or back up.



lp

Sends a file to the printer queue ("line printer").
lp filename.txt



tar

Bundles files/folders into one archive. tar -cvf archive.tar folder/ to create, -xvf to extract.



gzip

Compresses a file into a .gz version. Pair with tar for the classic .tar.gz.

PART 3

Your Turn: Practice on Ubuntu

Open a terminal and follow along — the core workflow from today, start to finish.

1

```
pwd && ls -la
```

See where you are and what's here

2

```
mkdir practice && cd practice
```

Make and enter a working folder

3

```
vi notes.txt
```

Insert mode, type a line, Esc, :wq

4

```
cp notes.txt backup.txt
```

Copy it, then cat and wc -l both

5

```
tar -cvf files.tar *.txt
```

Archive, then gzip files.tar to compress

6

```
rm backup.txt files.tar.gz
```

Clean up



```
$ man <any-command>
```

```
... keep exploring
```

```
$ echo "Thank you!"
```

Questions?

`man <command>` is always the fastest way to look something up.